CELL IDENTITAS

```
NAMA = 'Muhammad Pearl Ocshada'
NIM = '244107020064'
print(NAMA, NIM)
```

```
Muhammad Pearl Ocshada 244107020064
```

Mount Drive

```
import cv2
import matplotlib.pyplot as plt
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

1. GAMBAR PLAT NOMOR BINER

```
plat_no = cv2.imread("/content/drive/MyDrive/PCVK/Images/plat_nomer.jpg")

grey_plat = cv2.cvtColor(plat_no, cv2.COLOR_BGR2GRAY)

ref, plat_no = cv2.threshold(grey_plat, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)

plt.figure(figsize=(10,10))

plt.subplot(131)
plt.imshow(cv2.cvtColor(plat_no, cv2.COLOR_BGR2RGB))
plt.title('Citra Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(132)
plt.imshow(grey_plat, cmap='gray')
plt.title('Citra Grayscale')
plt.xticks([])
plt.yticks([])

plt.subplot(133)
plt.imshow(plat_no, cmap='gray')
plt.title('Citra Biner (Otsu)')
plt.xticks([])
plt.yticks([])

plt.show()
```



2. GAMBAR NOISY TEXT

```
img_noisy = cv2.imread("/content/drive/MyDrive/PCVK/Images/noisy_text.png")

grey_noisy = cv2.cvtColor(img_noisy, cv2.COLOR_BGR2GRAY)

ref, noisy_text = cv2.threshold(grey_noisy, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)

plt.figure(figsize=(10,10))

plt.subplot(131)
plt.imshow(cv2.cvtColor(img_noisy, cv2.COLOR_BGR2RGB))
plt.title('Citra Asli')
```

```
plt.xticks([])
plt.yticks([])

plt.subplot(132)
plt.imshow(grey_noisy, cmap='gray')
plt.title('Citra Grayscale')
plt.xticks([])
plt.yticks([])

plt.subplot(133)
plt.imshow(noisy_text, cmap='gray')
plt.title('Citra Biner (Otsu)')
plt.xticks([])
plt.yticks([])

plt.show()
```



PRAKTIKUM MORFOLOGI

3. **Fungsi Dilasi**

```
import numpy as np

def dilasi_citra(F,w):
  #size image
  p,q = F.shape

  imgD = np.zeros((p,q),dtype = np.uint8)

  SED = np.ones ((w,w),dtype=np.uint8)
  constant1 = (w-1)//2

  #Proses Dilasi
  for i in range(constant1, p-constant1):
    for j in range(constant1,q-constant1):
      temp= F[i-constant1:i+constant1+1, j-constant1:j+constant1+1]
      product= temp*SED
      imgD[i,j]= np.max(product)
  return imgD

img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

dilasi_noisy = dilasi_citra(img_noisy_text,5)
dilasi_plat = dilasi_citra(img_plat_no,5)

plt.figure(figsize=(10,10))

#Menampilkan 4 gambar Citra asli dan Dilasi
plt.figure(figsize=(10,10))

plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,2)
plt.imshow(dilasi_noisy, cmap='gray')
plt.title('Hasil Dilasi Noisy Text')
plt.xticks([])
plt.yticks([])

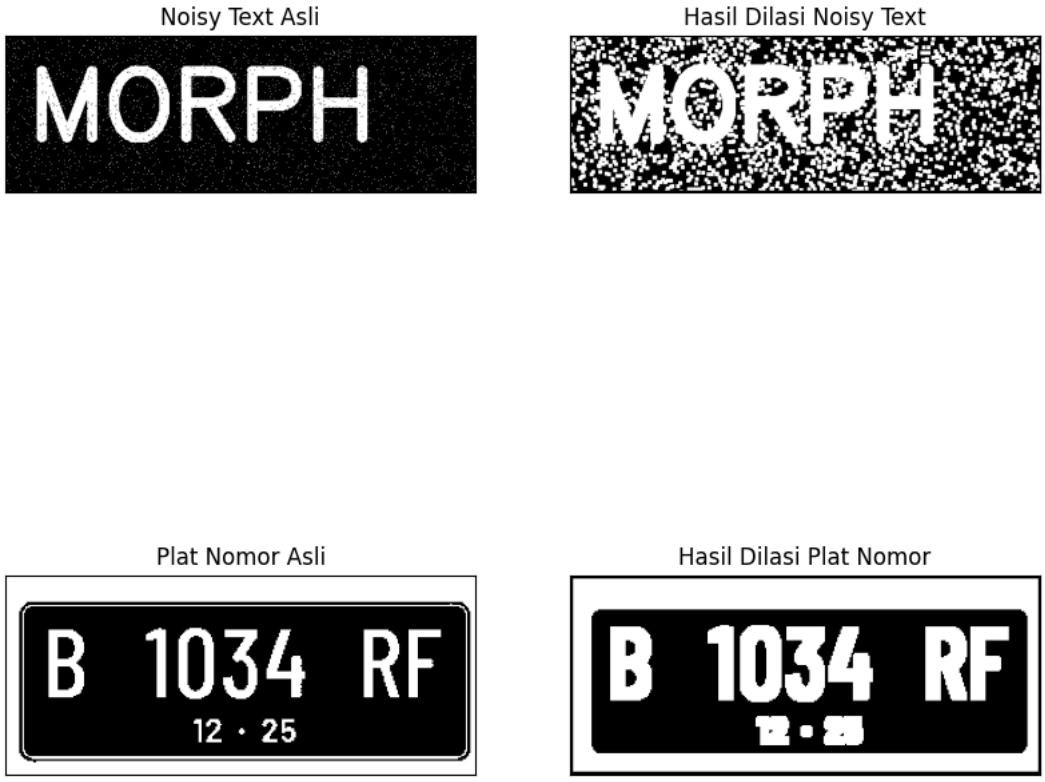
plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
```

```
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Dilasi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
<Figure size 1000x1000 with 0 Axes>
```



4. Implementasi Dilasi OpenCV

```
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

kernel = np.ones((3, 3), np.uint16)

dilasi_noisy = cv2.dilate(img_noisy_text, kernel, iterations=1)
dilasi_plat = cv2.dilate(img_plat_no, kernel, iterations=1)

plt.figure(figsize=(10,10))

plt.subplot(2, 2, 1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

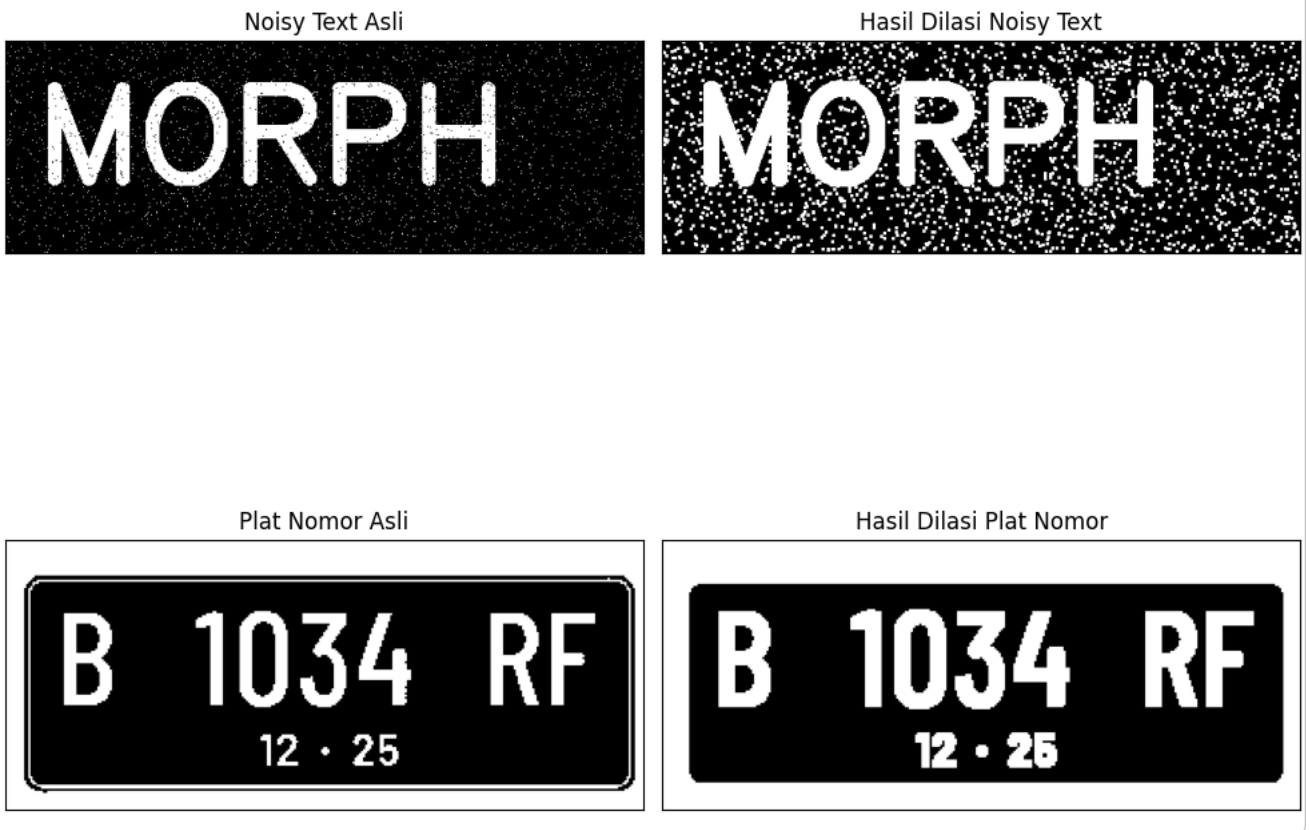
plt.subplot(2, 2, 2)
plt.imshow(dilasi_noisy, cmap='gray')
plt.title('Hasil Dilasi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2, 2, 3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2, 2, 4)
```

```
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Dilasi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.tight_layout()
plt.show()
```



5. Fungsi Erosi

```
def erosi_citra(F,w):
  #size image
  m,n = F.shape

  SE = np.ones((w,w),dtype=np.uint8)
  constant2 = (w-1)//2
  imgE = np.zeros((m,n),dtype=np.uint8)

  #proses erosi
  for i in range(constant2, m-constant2):
    for j in range(constant2,n-constant2):
      temp= F[i-constant2:i+constant2+1, j-constant2:j+constant2+1]
      product= temp*SE
      imgE[i,j]= np.min(product)
  return imgE

img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

erosi_noisy = erosi_citra(img_noisy_text,5)
erosi_plat = erosi_citra(img_plat_no,5)

plt.figure(figsize=(10,10))

#Menampilkan 4 gambar Citra asli dan Erosi
plt.figure(figsize=(10,10))

plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
```

```
plt.yticks([])

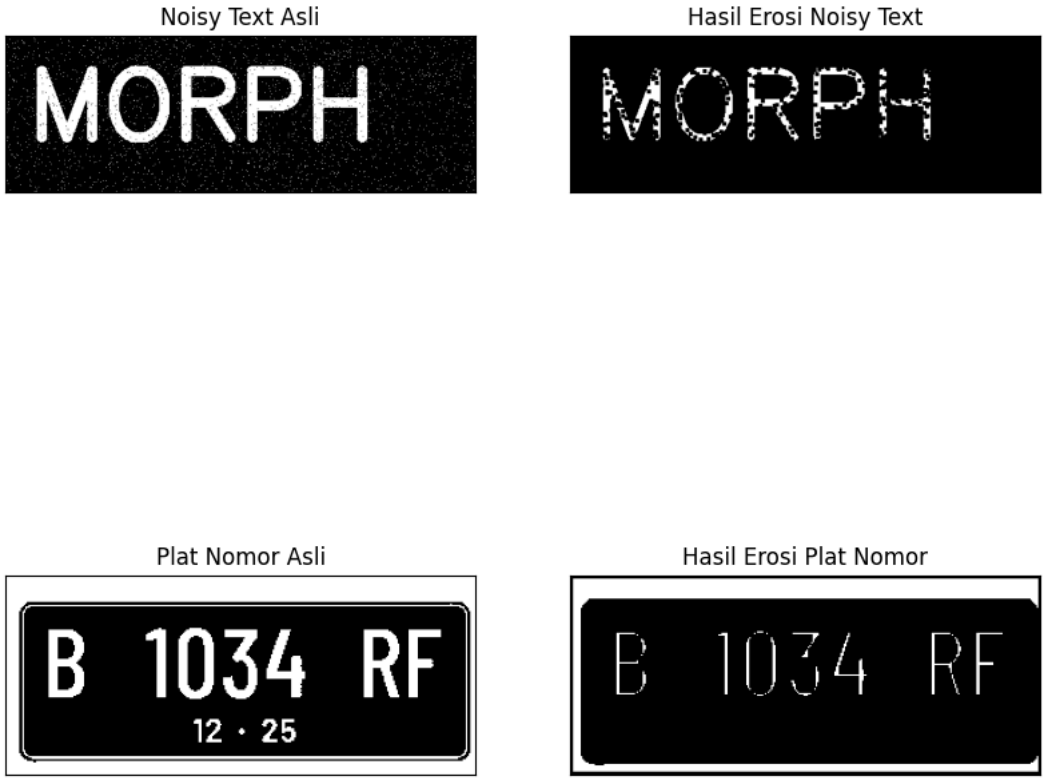
plt.subplot(2,2,2)
plt.imshow(erosi_noisy, cmap='gray')
plt.title('Hasil Erosi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(erosi_plat, cmap='gray')
plt.title('Hasil Erosi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
<Figure size 1000x1000 with 0 Axes>
```



6. Implementasi Erosi OpenCV

```
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

kernel = np.ones((3, 3), np.uint16)

erosi_noisy = cv2.erode(img_noisy_text, kernel, iterations=1)
erosi_plat = cv2.erode(img_plat_no, kernel, iterations=1)

plt.figure(figsize=(10,10))

plt.subplot(2, 2, 1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

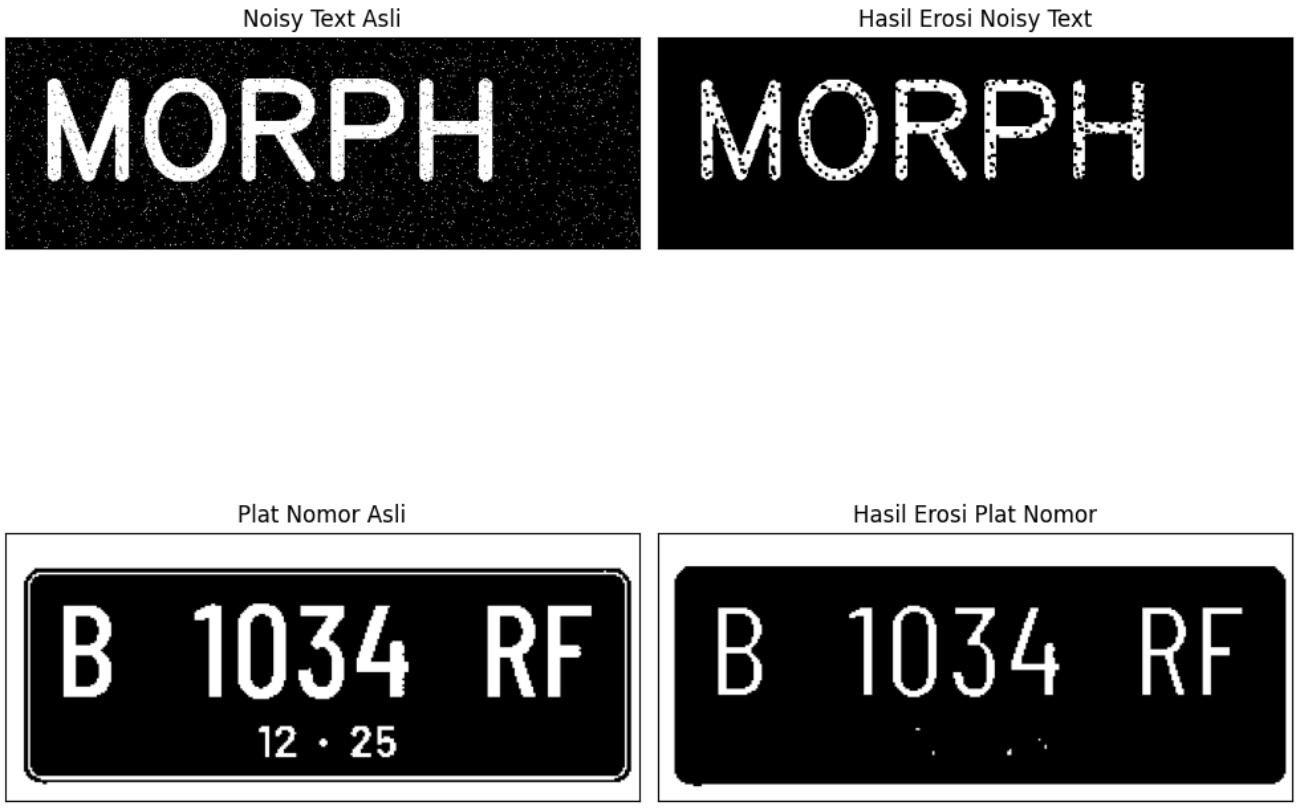
plt.subplot(2, 2, 2)
plt.imshow(erosi_noisy, cmap='gray')
```

```
plt.title('Hasil Erosi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2, 2, 3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2, 2, 4)
plt.imshow(erosi_plat, cmap='gray')
plt.title('Hasil Erosi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.tight_layout()
plt.show()
```



7. Operasi Opening

- Tanpa Library

```
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

erosi_noisy = erosi_citra(img_noisy_text,5)
dilasi_noisy = dilasi_citra(erosi_noisy,5)

erosi_plat = erosi_citra(img_plat_no,5)
dilasi_plat = dilasi_citra(erosi_plat,5)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

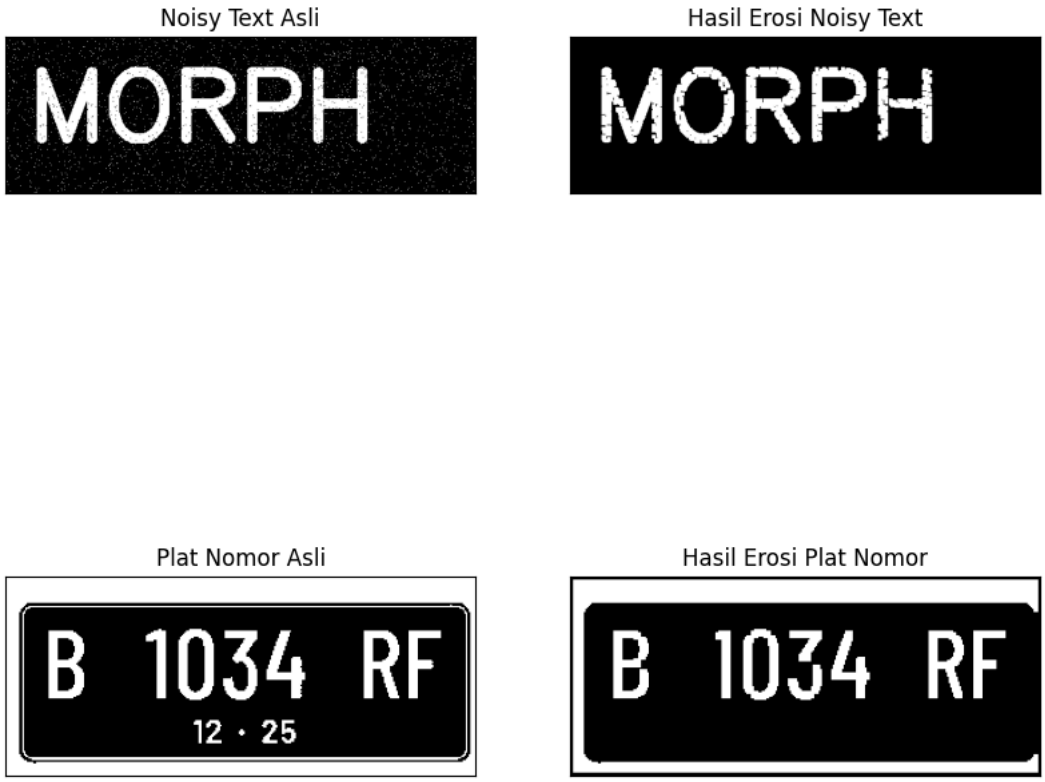
plt.subplot(2,2,2)
plt.imshow(dilasi_noisy, cmap='gray')
```

```
plt.title('Hasil Erosi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Erosi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```



- Menggunakan Library Morphology

```
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

kernel = np.ones((3, 3), np.uint16)

opening_noisy = cv2.morphologyEx(img_noisy_text, cv2.MORPH_OPEN, kernel)
opening_plat = cv2.morphologyEx(img_plat_no, cv2.MORPH_OPEN,kernel)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

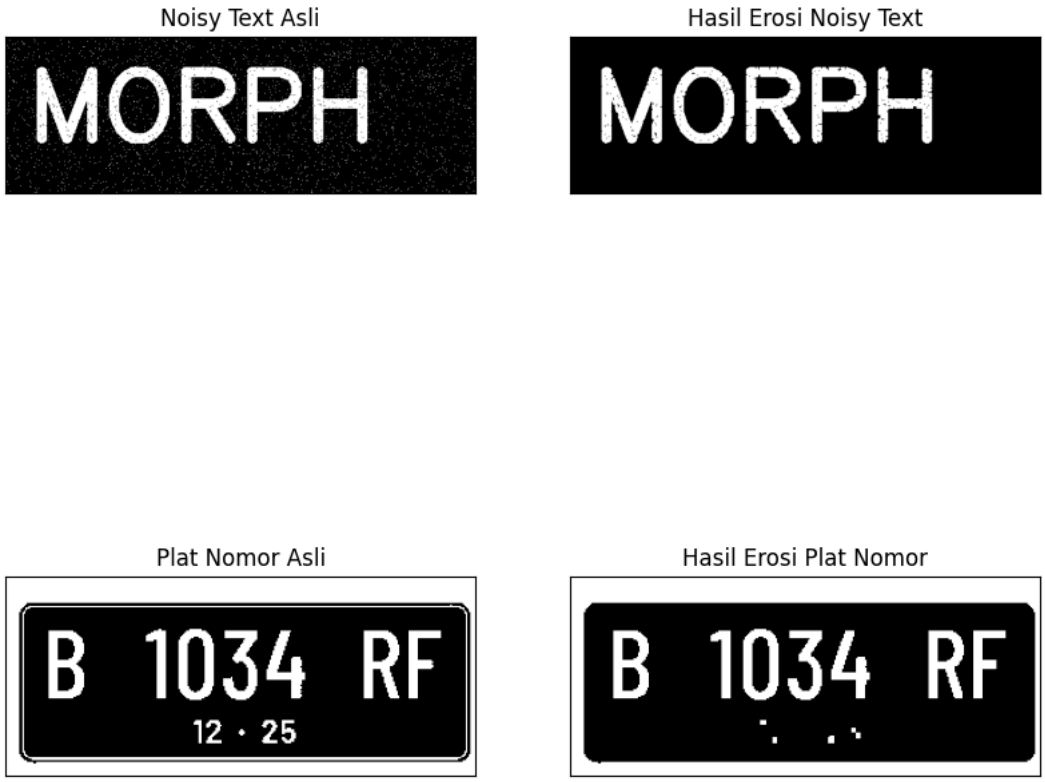
plt.subplot(2,2,2)
plt.imshow(opening_noisy, cmap='gray')
plt.title('Hasil Erosi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
```

```
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(opening_plat, cmap='gray')
plt.title('Hasil Erosi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```



8. Operasi Closing

- Tanpa library

```
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

closing_noisy = erosi_citra(dilasi_citra(img_noisy_text,7),7)
closing_plat = erosi_citra(dilasi_citra(img_plat_no,7),7)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

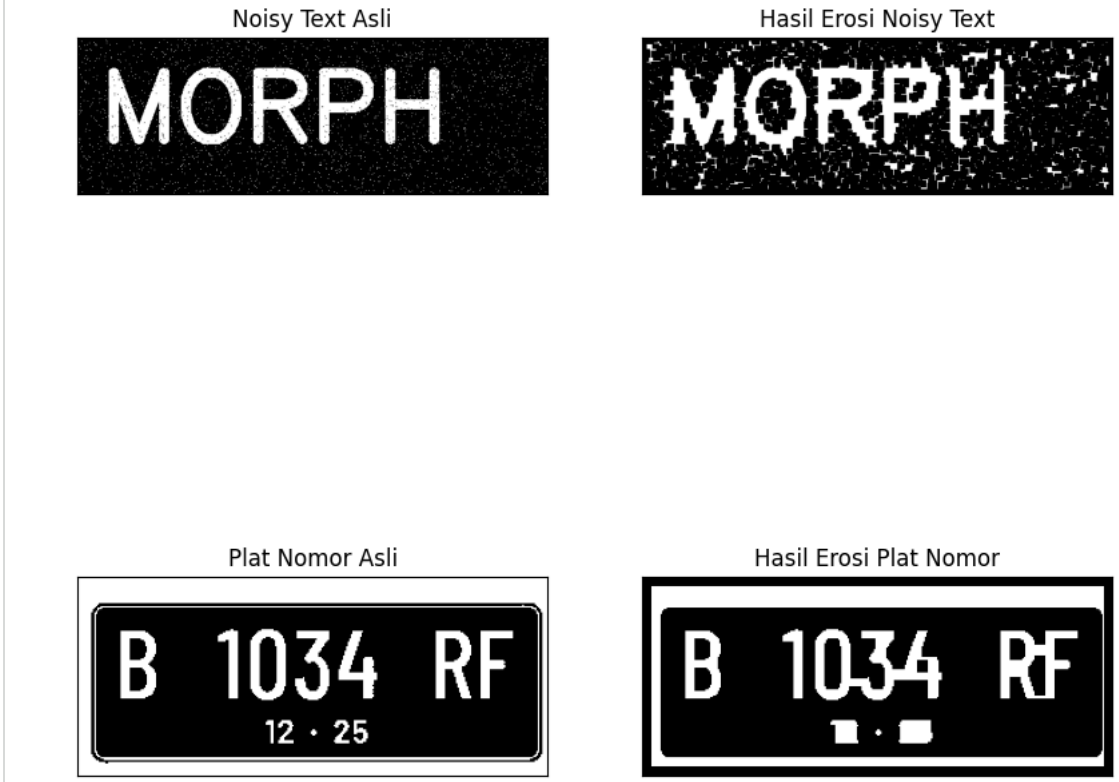
plt.subplot(2,2,2)
plt.imshow(closing_noisy, cmap='gray')
plt.title('Hasil Erosi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
```

```
plt.imshow(closing_plat, cmap='gray')
plt.title('Hasil Erosi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```



- Menggunakan Library Morphology

```
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

kernel = np.ones((3, 3), np.uint8)

closing_noisy = cv2.morphologyEx(img_noisy_text, cv2.MORPH_CLOSE, kernel)
closing_plat = cv2.morphologyEx(img_plat_no, cv2.MORPH_CLOSE,kernel)

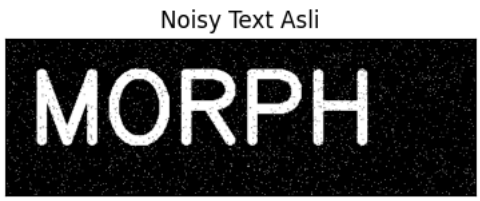
plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,2)
plt.imshow(closing_noisy, cmap='gray')
plt.title('Hasil Erosi Noisy Text')
plt.xticks([])
plt.yticks([])

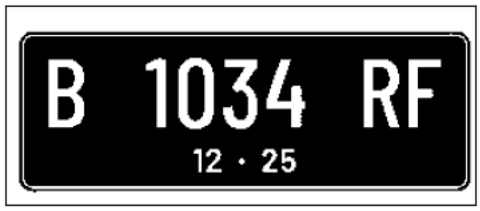
plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

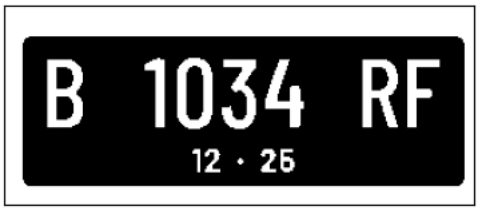
plt.subplot(2,2,4)
plt.imshow(closing_plat, cmap='gray')
plt.title('Hasil Erosi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```









9. Operasi Dilasi berbentuk Cross 3 x 3 dan 5 x 5

- Cross 3x3

```
def dilasi_citra(F,w):
  p,q = F.shape

  imgD = np.zeros((p,q),dtype = np.uint8)

  SED = np.array([[0, 1, 0],
                  [1, 1, 1],
                  [0, 1, 0]], dtype = np.uint8)
  constant1 = (w-1)//2

  for i in range(constant1, p-constant1):
    for j in range(constant1, q-constant1):
      temp= F[i-constant1:i+constant1+1, j-constant1:j+constant1+1]
      product= temp*SED
      imgD[i,j]= np.max(product)
  return imgD

print('SED = CROSS 3')
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

dilasi_noisy = dilasi_citra(img_noisy_text,3)
dilasi_plat = dilasi_citra(img_plat_no,3)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,2)
plt.imshow(dilasi_noisy, cmap='gray')
plt.title('Hasil Dilasi Noisy Text')
plt.xticks([])
plt.yticks([])

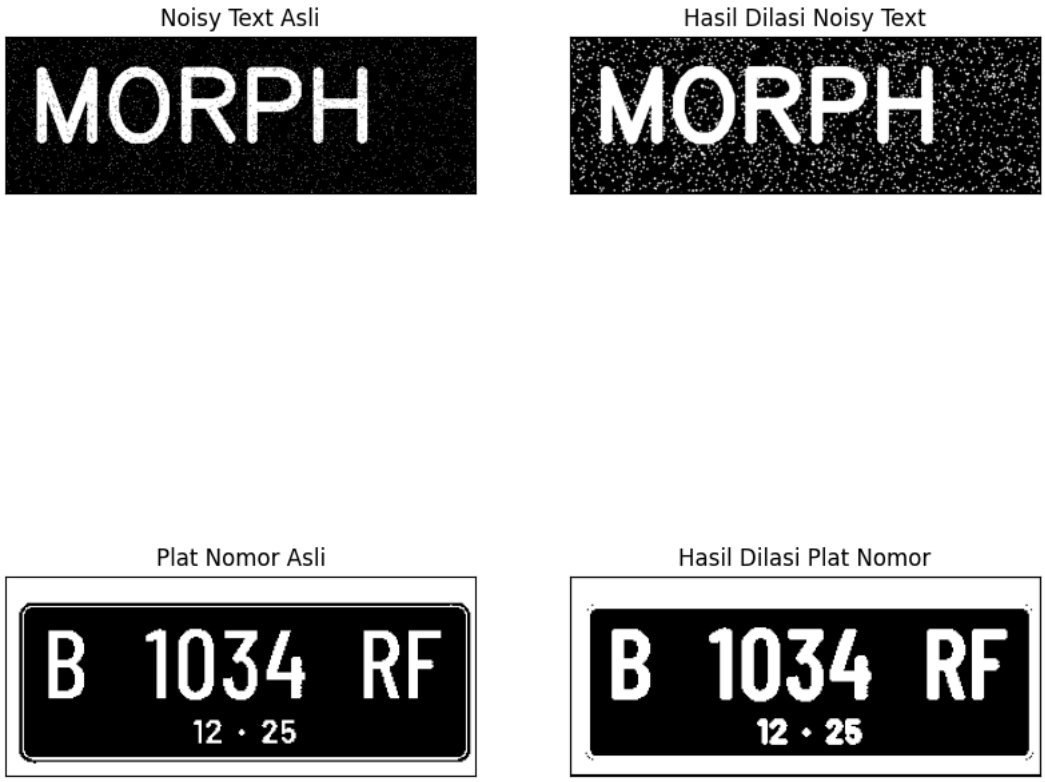
plt.subplot(2,2,3)
```

```
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Dilasi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
SED = CROSS 3
```



- Cross 5x5

```
def dilasi_citra(F,w):
  p,q = F.shape

  imgD = np.zeros((p,q),dtype = np.uint8)

  # Structuring Element berbentuk Cross 5x5 yang simetris
  SED = np.array([[0, 0, 1, 0, 0],
                  [0, 0, 1, 0, 0],
                  [1, 1, 1, 1, 1],
                  [0, 0, 1, 0, 0],
                  [0, 0, 1, 0, 0]], dtype = np.uint8)
  constant1 = (w-1)//2

  for i in range(constant1, p-constant1):
    for j in range(constant1, q-constant1):
      temp= F[i-constant1:i+constant1+1, j-constant1:j+constant1+1]
      product= temp*SED
      imgD[i,j]= np.max(product)
  return imgD

# Memulihkan fungsi print bawaan dan memanggilnya dengan benar
import builtins
print = builtins.print
print('SED = CROSS 5')

img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()
```

```
dilasi_noisy = dilasi_citra(img_noisy_text,5)
dilasi_plat = dilasi_citra(img_plat_no,5)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

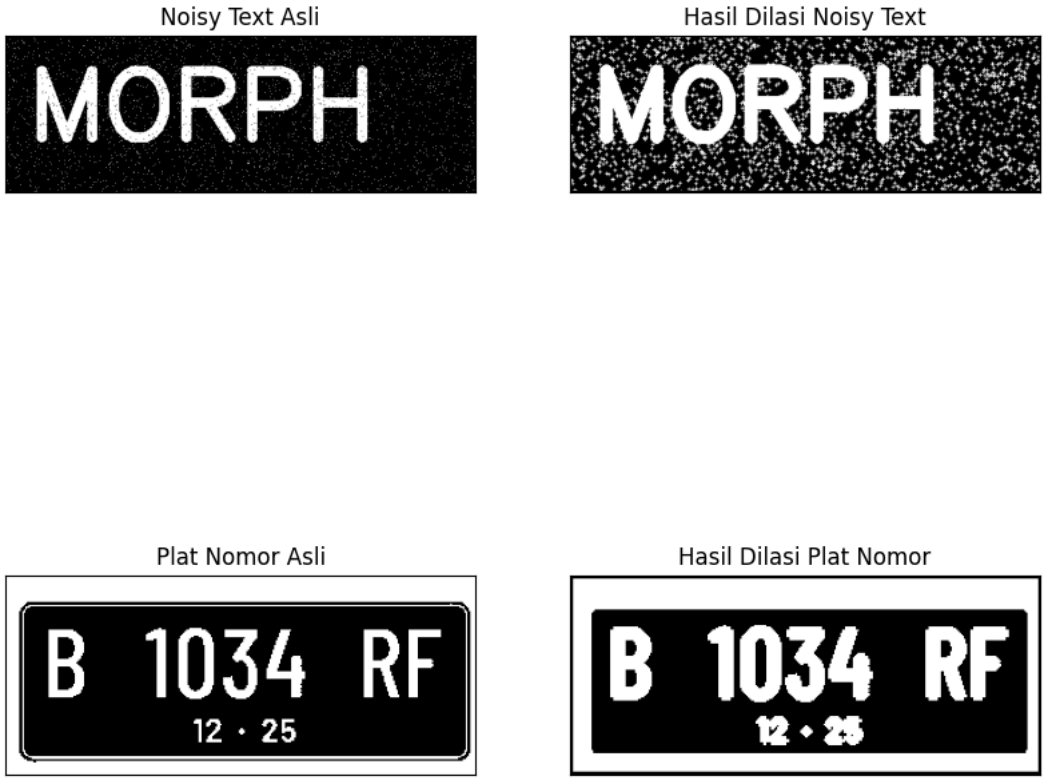
plt.subplot(2,2,2)
plt.imshow(dilasi_noisy, cmap='gray')
plt.title('Hasil Dilasi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Dilasi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
SED = CROSS 5
```



10. Operasi dilasi circular 3x3 dan 5x5

- dilasi circular 3x3

```
def dilasi_citra(F,w):
  p,q = F.shape

  imgD = np.zeros((p,q),dtype = np.uint8)

  SED = cv2.getStructuringElement(cv2.MORPH_ELLIPSE,(w,w))
  constant1 = (w-1)//2
```

```
  for i in range(constant1, p-constant1):
    for j in range(constant1, q-constant1):
      temp= F[i-constant1:i+constant1+1, j-constant1:j+constant1+1]
      product= temp*SED
      imgD[i,j]= np.max(product)
  return imgD

print('SED = CIRCULAR 3')
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

dilasi_noisy = dilasi_citra(img_noisy_text,3)
dilasi_plat = dilasi_citra(img_plat_no,3)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

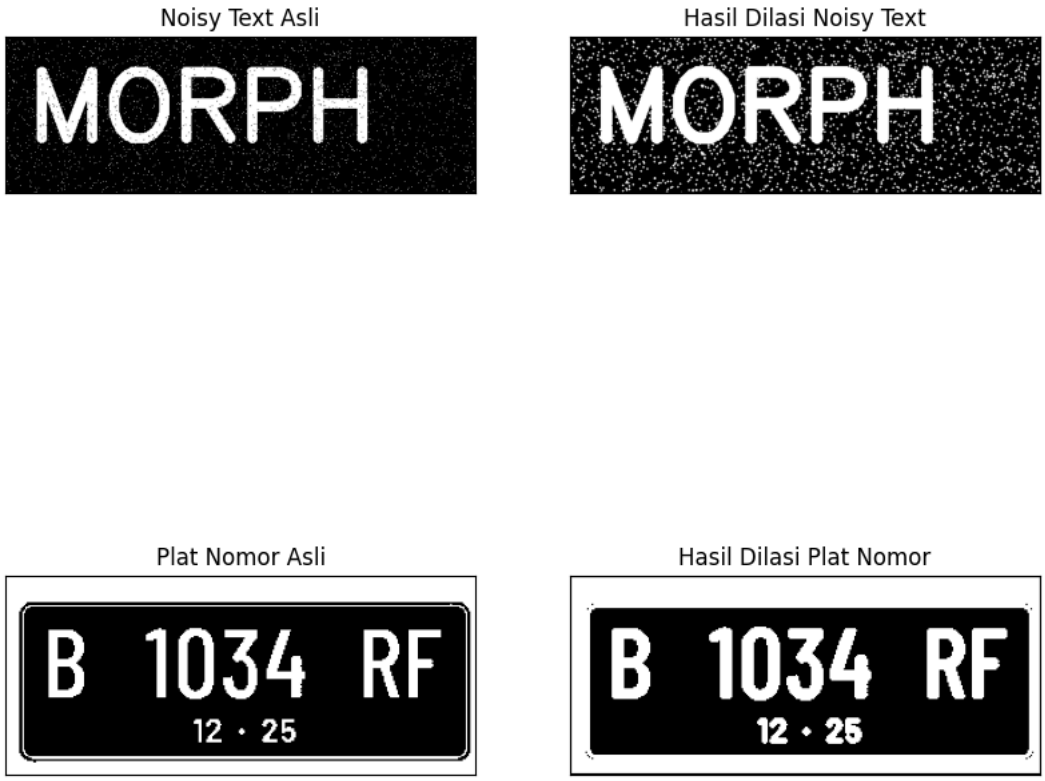
plt.subplot(2,2,2)
plt.imshow(dilasi_noisy, cmap='gray')
plt.title('Hasil Dilasi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Dilasi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
SED = CIRCULAR 3
```



- dilasi circular 5x5

```
def dilasi_citra(F,w):
  p,q = F.shape

  imgD = np.zeros((p,q),dtype = np.uint8)

  SED = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (w, w))
  constant1 = (w-1)//2

  for i in range(constant1, p-constant1):
    for j in range(constant1, q-constant1):
      temp= F[i-constant1:i+constant1+1, j-constant1:j+constant1+1]
      product= temp*SED
      imgD[i,j]= np.max(product)
  return imgD

print('SED = CIRCULAR 5')
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

dilasi_noisy = dilasi_citra(img_noisy_text,5)
dilasi_plat = dilasi_citra(img_plat_no,5)

plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

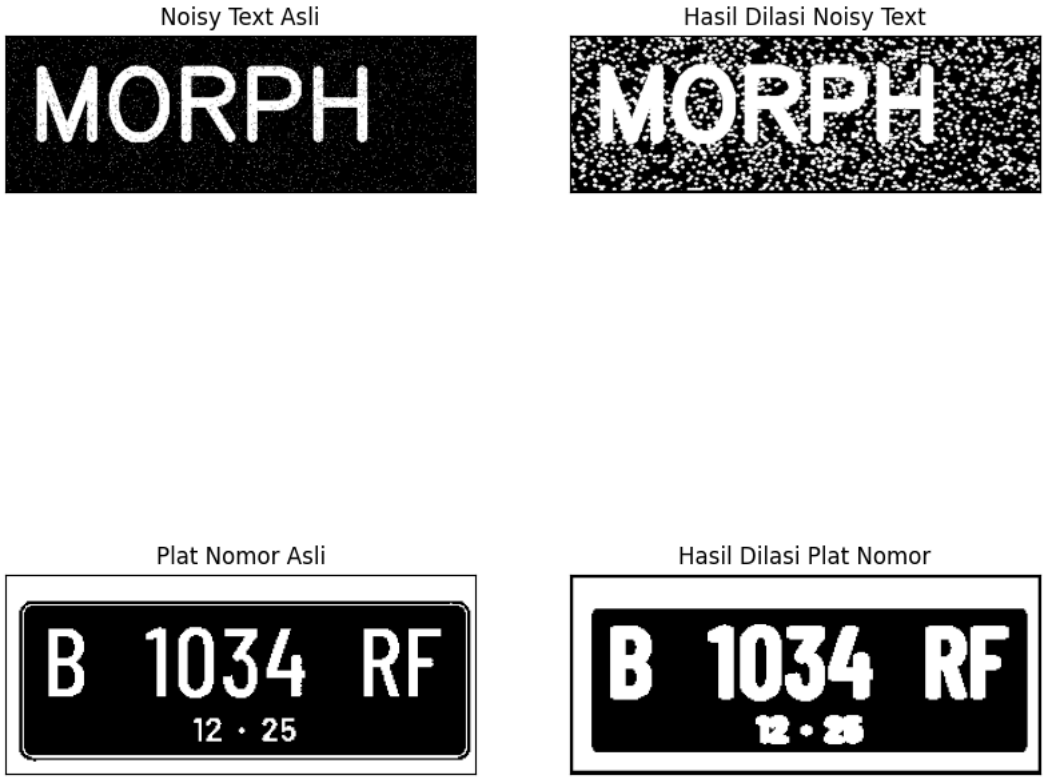
plt.subplot(2,2,2)
plt.imshow(dilasi_noisy, cmap='gray')
plt.title('Hasil Dilasi Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(2,2,4)
plt.imshow(dilasi_plat, cmap='gray')
plt.title('Hasil Dilasi Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
SED = CIRCULAR 5
```



11. Operasi Erosi dengan Structuring Element Rectangle 3 x 5 dan 5 x 7

- Erosi Rectangle 3x5 & 5x7

```
def erosi_rectangle(F, p, l):
  m, n = F.shape

  # Sesuaikan urutan dimensi (lebar, tinggi) di OpenCV agar menghasilkan matriks berukuran (p, l) di NumPy
  SE = cv2.getStructuringElement(cv2.MORPH_RECT, (l, p))
  constant1 = (p-1)//2
  constant2 = (l-1)//2

  imgE = np.zeros((m, n), dtype=np.uint8)

  for i in range(constant1, m-constant1):
    for j in range(constant2, n-constant2):
      temp = F[i-constant1:i+constant1+1, j-constant2:j+constant2+1]
      product = temp * SE
      imgE[i, j] = np.min(product)

  return imgE

print('SE = RECTANGLE 3x5')
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

erosi_noisy = erosi_rectangle(img_noisy_text, 3, 5)
erosi_plat = erosi_rectangle(img_plat_no, 3, 5)

print('SE = RECTANGLE 5x7')
erosi_noisy_2 = erosi_rectangle(img_noisy_text, 5, 7)
erosi_plat_2 = erosi_rectangle(img_plat_no, 5, 7)

# Menggunakan grid 4x2 agar dapat menampung seluruh 8 subplot
plt.figure(figsize=(10, 14))

# Baris 1: Erosi Rect 3x5 (Noisy Text)
plt.subplot(4, 2, 1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
```

```
plt.yticks([])

plt.subplot(4, 2, 2)
plt.imshow(erosi_noisy, cmap='gray')
plt.title('Hasil Erosi Rect 3x5 Noisy Text')
plt.xticks([])
plt.yticks([])

# Baris 2: Erosi Rect 3x5 (Plat Nomor)
plt.subplot(4, 2, 3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4, 2, 4)
plt.imshow(erosi_plat, cmap='gray')
plt.title('Hasil Erosi Rect 3x5 Plat Nomor')
plt.xticks([])
plt.yticks([])

# Baris 3: Erosi Rect 5x7 (Noisy Text)
plt.subplot(4, 2, 5)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4, 2, 6)
plt.imshow(erosi_noisy_2, cmap='gray')
plt.title('Hasil Erosi Rect 5x7 Noisy Text')
plt.xticks([])
plt.yticks([])

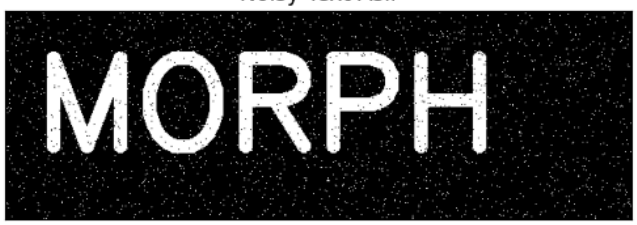
# Baris 4: Erosi Rect 5x7 (Plat Nomor)
plt.subplot(4, 2, 7)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4, 2, 8)
plt.imshow(erosi_plat_2, cmap='gray')
plt.title('Hasil Erosi Rect 5x7 Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.tight_layout()
plt.show()
```

```
SE = RECTANGLE 3x5
SE = RECTANGLE 5x7
```

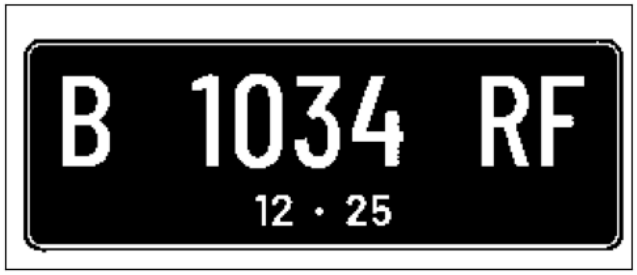
Noisy Text Asli



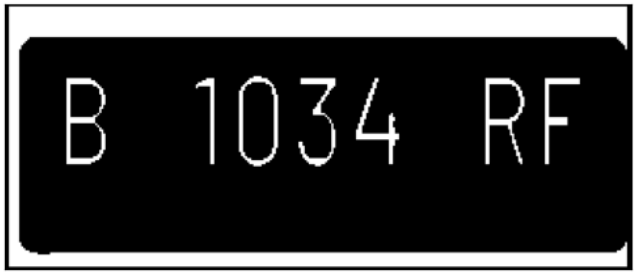
Hasil Erosi Rect 3x5 Noisy Text



Plat Nomor Asli



Hasil Erosi Rect 3x5 Plat Nomor



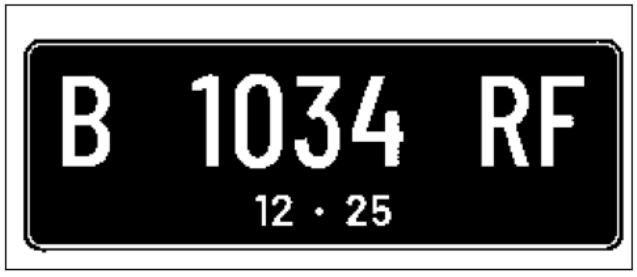
Noisy Text Asli



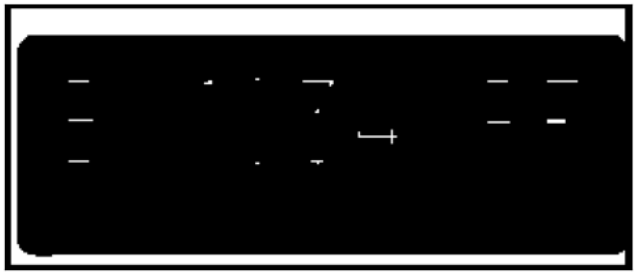
Hasil Erosi Rect 5x7 Noisy Text



Plat Nomor Asli



Hasil Erosi Rect 5x7 Plat Nomor



12. Erosi Line Vertikal 3 & 5

```
def erosi_citra(F,k):
  m,n = F.shape

  # Sesuaikan urutan dimensi (lebar, tinggi) di OpenCV agar menghasilkan matriks berukuran (p, l) di NumPy
  SE = cv2.getStructuringElement(cv2.MORPH_ELLIPSE,(1,k))
  constant1 = (k-1)//2
```

```
  imgE = np.zeros((m,n),dtype = np.uint8)

  for i in range(constant1, m-constant1):
    for j in range(constant1, n-constant1):
      temp = F[i-constant1:i+constant1+1, j-constant1:j+constant1+1]
      product = temp * SE
      imgE[i,j] = np.min(product)

  return imgE

print('SE = LINE VERTIKAL 3')
img_noisy_text = noisy_text.copy()
img_plat_no = plat_no.copy()

erosi_noisy = erosi_citra(img_noisy_text,3)
erosi_plat = erosi_citra(img_plat_no,3)

print('SE = LINE VERTIKAL 5')
erosi_noisy_2 = erosi_citra(img_noisy_text,5)
erosi_plat_2 = erosi_citra(img_plat_no,5)

plt.figure(figsize=(10,10))
plt.subplot(4,2,1)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,2)
plt.imshow(erosi_noisy, cmap='gray')
plt.title('Hasil Erosi Line Vertikal 3 Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,3)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,4)
plt.imshow(erosi_plat, cmap='gray')
plt.title('Hasil Erosi Line Vertikal 3 Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,5)
plt.imshow(img_noisy_text, cmap='gray')
plt.title('Noisy Text Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,6)
plt.imshow(erosi_noisy_2, cmap='gray')
plt.title('Hasil Erosi Line Vertikal 5 Noisy Text')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,7)
plt.imshow(img_plat_no, cmap='gray')
plt.title('Plat Nomor Asli')
plt.xticks([])
plt.yticks([])

plt.subplot(4,2,8)
plt.imshow(erosi_plat_2, cmap='gray')
plt.title('Hasil Erosi Line Vertikal 5 Plat Nomor')
plt.xticks([])
plt.yticks([])

plt.show()
```

```
SE = LINE VERTIKAL 3
SE = LINE VERTIKAL 5
```



13. Transformasi TOP-HAT element 3x3

```
filtersize = (3,3)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, filtersize)

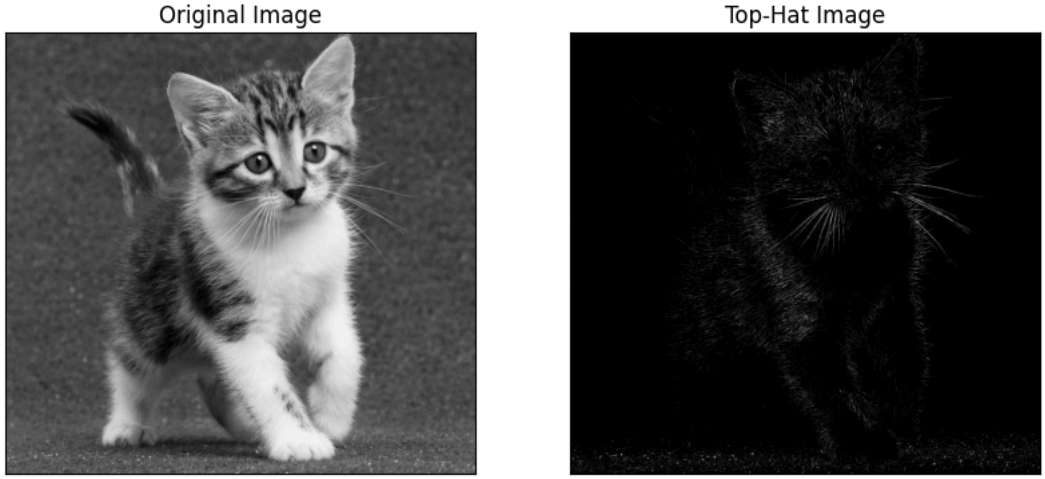
img = cv2.imread('/content/drive/MyDrive/PCVK/Images/kitten01 (1).jpg')
img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

tophat_img = cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel)

plt.figure(figsize=(10,10))
plt.subplot(1,2,1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.xticks([])
plt.yticks([])

plt.subplot(1,2,2)
plt.imshow(tophat_img, cmap='gray')
plt.title('Top-Hat Image')
plt.xticks([])
plt.yticks([])

plt.show()
```



14. Transformasi Black-Hat element 3x3

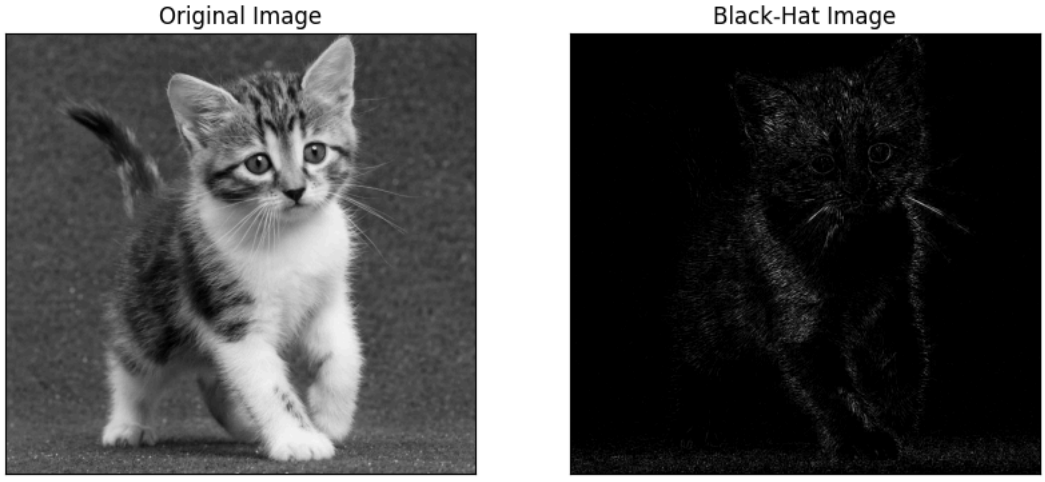
```
kitten = img.copy()
filtersize = (3,3)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, filtersize)

blackhat_img = cv2.morphologyEx(kitten, cv2.MORPH_BLACKHAT,kernel)

plt.figure(figsize=(10,10))
plt.subplot(1,2,1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.xticks([])
plt.yticks([])

plt.subplot(1,2,2)
plt.imshow(blackhat_img, cmap='gray')
plt.title('Black-Hat Image')
plt.xticks([])
plt.yticks([])

plt.show()
```



15. Program skeleton pada citra biner

```
import cv2 as cv
from matplotlib import pyplot as plt
from skimage import filters
from skimage.morphology import skeletonize

# Read image in grayscale mode
```

```
lily_img = cv.imread('/content/drive/MyDrive/PCVK/Images/lily.jpg', cv.IMREAD_GRAYSCALE)

binary = lily_img > filters.threshold_triangle(lily_img)

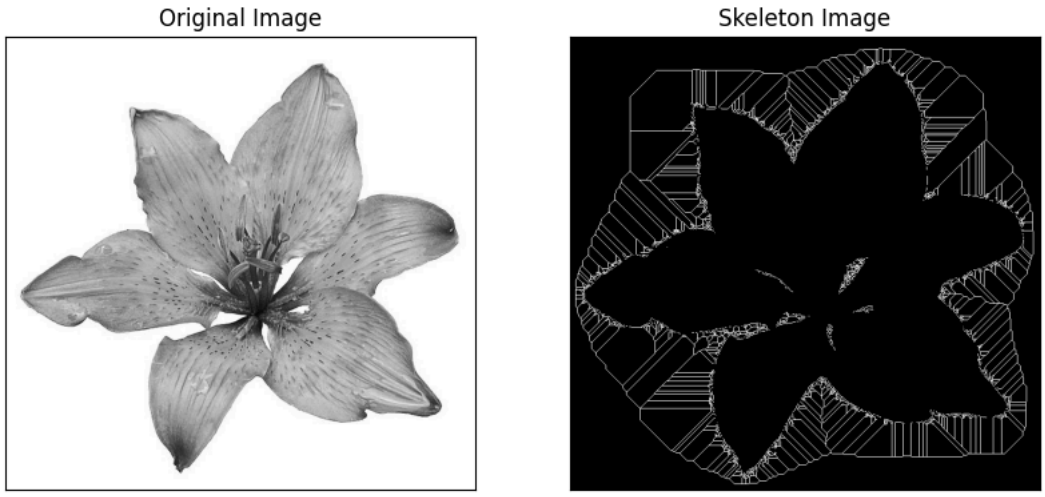
binary_cp = binary.copy()
binary_cp[binary_cp == True] = 1
binary_cp[binary_cp == False] = 0

skeleton = skeletonize(binary_cp)

plt.figure(figsize=(10,10))
plt.subplot(1,2,1)
plt.imshow(lily_img, cmap='gray')
plt.title('Original Image')
plt.xticks([])
plt.yticks([])

plt.subplot(1,2,2)
plt.imshow(skeleton, cmap='gray')
plt.title('Skeleton Image')
plt.xticks([])
plt.yticks([])

plt.savefig("binary and bin_skeleton.png")
plt.show()
```



16. Program skeleton pada citra hasil inverse

```
from skimage import io
from matplotlib import pyplot as plt
from skimage import filters
from skimage.morphology import skeletonize
import cv2 as cv

img = cv.imread('/content/drive/MyDrive/PCVK/Images/lily.jpg')[...,0]
img1 = skeleton.copy()

binary = img > filters.threshold_triangle(img)

skeleton1 = skeletonize(binary)

# Create a 2x2 subplot grid to accommodate 4 images
fig, ax = plt.subplots(2, 2, figsize=(10, 10))

ax[0, 0].imshow(img, cmap='gray')
ax[0, 0].set_title('Original Image')
ax[0, 0].axis('off')

ax[0, 1].imshow(img1, cmap='gray')
ax[0, 1].set_title('Skeleton Image Langkah 15')
ax[0, 1].axis('off')

ax[1, 0].imshow(img, cmap='gray')
ax[1, 0].set_title('Original Image')
```

```
ax[1, 0].axis('off')

ax[1, 1].imshow(skeleton1, cmap='gray')
ax[1, 1].set_title('Skeleton Image')
ax[1, 1].axis('off')

plt.tight_layout()
plt.show()
```

